**LESSON ONE**

**DEFINITION OF TERMS**

**Data**

Data are raw, unprocessed facts, figures, or symbols that have little or no meaning on their own. They can be numbers, text, images, or observations. Data are the **input** for creating information. For an example "50, 70, 85" 'n just numbers without context.

**Information**

Information is data that has been processed, organized, or structured in a way that gives it meaning and usefulness. It answers questions like *who, what, where, when, and how.* It supports decision-making. For example "Students scored 50, 70, and 85 in LIS 327 test" 'n now meaningful

**Information Management**

Information Management is the systematic process of collecting, organizing, storing, retrieving, and disseminating information for effective use. It ensures information is accessible, accurate, and timely.

**Key Processes Include:**

- Creation/collection
- Organization (classification, cataloguing)
- Storage

- Retrieval
- Dissemination

**Information Products**

Library and information products encompass a diverse range of resources that serve the information needs of users in libraries and other information settings. These products are essential for research, education, entertainment, and personal enrichment. They can be classified into various categories based on their format and purpose. Some common categories include documents such as periodicals, reference materials like encyclopedias and dictionaries, books in different genres and formats, multimedia materials like videos and DVDs, government publications, corporate reports, patents, conference proceedings, and various services such as information services, document delivery, and training. Each category of library and information products plays a significant role in facilitating access to knowledge, supporting research and learning, and fulfilling the diverse information requirements of users.

**Range of Information Product**

**Documentary information products**,

These include traditional print and recorded materials. These are among the oldest forms of information products and consist of books, journals, newspapers, reports, and theses. They are usually well-structured and serve as authoritative sources of information for academic and research purposes. For example, textbooks used in universities and scholarly journals containing research articles fall under this category.

**Bibliographic information products**

They are also significant in the field of library and information science. These products do not necessarily contain full information but instead provide details about other infor

mation sources. Examples include bibliographies, indexes, abstracts, and catalogues. Tools like the Online Public Access Catalogue (OPAC) serve as bibliographic products that help users identify and locate materials within a library collection.

**Electronic or digital information products**

Electronic or digital information products **are** increasingly prominent due to technological advancements. These include e-books, e-journals, online databases, institutional repositories, and multimedia resources. Platforms such as an Institutional Repository store and provide access to digital scholarly works. These products offer advantages such as remote access, easy updating, and wide dissemination.

**Value-added information products**

These information products are created by analyzing, summarizing, or repackaging existing information to make it more useful. Examples include children fiction, research reports, policy briefs, newsletters, and review articles e.t.c. These products interpret and present information in a way that supports decision-making and saves users time.

**Multimedia information products**

This types of information products combine different forms of content such as text, audio, video, and graphics to convey information. These include educational videos, podcasts, and interactive learning materials. Multimedia products are especially effective in enhancing understanding and engagement, particularly in digital learning environments.

**Institutional and organizational information products**

These are produced by institutions to communicate specific information about their activities, services, or performance. These include annual reports, strategic plans, bulletins, and newsletters. Such products are important for accountability, communication, and org

anizational visibility.

**Information Services**

Information services refer to the various activities, functions, and assistance provided by libraries and information centres to help users identify, access, retrieve, and effectively us e information.

In simple terms, while information products are *what* is provided (e.g., books, database s), information services are *how* users are helped to find and use those products. Inform ation services are intangible, they cannot be touched, seen but they can be felt. When co nsumers use the services they have feelings that something has been added to them esp ecially in solving their information needs

**Types of Information Services**

Information services in libraries and information centres are diverse and are designed to meet the varying needs of users by facilitating access to relevant information.

**1. Reference Services**

Reference service is one of the most fundamental information services provided in librari es. It involves personal assistance given by librarians to users in locating needed inform ation. Librarians help users identify sources, answer queries, and guide them in using libr ary tools such as catalogues, indexes, encyclopedias, and databases. This service enhan ces users' information searching skills and promotes effective information utilization.

Reference services may be:

- Ready reference service, which provides quick factual answers.
- Long-range reference service, which involves detailed research assistance.

- Digital or virtual reference service, delivered through email, chat, or online platforms.

**2. Current Awareness Service (CAS)**

Current Awareness Service keeps users informed about newly available information relevant to their interests or fields of study. Libraries regularly notify users about new books, journal issues, research developments, conferences, and institutional publications. The goal is to ensure that users remain up-to-date in their disciplines.

CAS may include:

- New arrival lists
- Bulletin boards
- Email alerts
- Library newsletters
- Social media updates

**3. Selective Dissemination of Information (SDI)**

Selective Dissemination of Information is a personalized information service designed to meet the specific needs of individual users. The librarian develops user profiles based on research interests and periodically sends relevant information materials to them.

SDI is particularly useful for researchers, lecturers, and postgraduate students who require specialized and continuous information updates.

**4. User Education and Information Literacy Services**

Libraries provide instructional programme aimed at teaching users how to effectively locate, evaluate, and use information resources. This service helps users become independent information seekers. Information literacy services are essential in combating misinformation and promoting academic integrity.

Activities include:

- Library orientation programmes
- Information literacy training
- Database searching instruction
- Research skills workshops
- Citation and referencing training

**5. Circulation Services**

Circulation service refers to activities related to lending and borrowing library materials. It ensures equitable access to library resources through systematic control of materials. Although this is administrative in nature, circulation service directly supports information access.

Functions include:

- Registration of users
- Charging and discharging materials
- Renewal and reservation of items

- Overdue notices and fines management

**6. Document Delivery and Interlibrary Loan Services**

When required information resources are not available locally, libraries obtain materials from other libraries or information centres. Interlibrary loan and document delivery services expand access beyond a single library's collection. Document Delivery and Interlibrary Loan Services promotes resource sharing and cooperation among libraries.

Users may receive:

- Photocopies of articles
- Scanned documents
- Borrowed books from partner institutions

**7. Bibliographic and Indexing Services**

Libraries prepare bibliographies, indexes, abstracts, and reading lists to guide users toward relevant information sources. These tools help users identify existing literature on particular subjects efficiently.

Examples include:

- Subject bibliographies
- Abstracting services
- Citation databases
- Institutional publication lists

**8. Referral Services**

Referral service involves directing users to other information agencies, experts, or institut ions when the required information is unavailable in the library. The librarian acts as an i nformation intermediary linking users with external knowledge sources.

**9. Digital and Electronic Information Services**

With the advancement of Information and Communication Technology (ICT), libraries no w provide electronic information services. These services allow users to access informati on anytime and from any location. They include;

- Online Public Access Catalogue (OPAC)
- Electronic databases
- E-books and e-journals access
- Institutional repositories
- Remote library access systems
- Virtual reference services

**10. Reprographic and Multimedia Services**

Libraries provide facilities for reproducing information materials in alternative formats. T hese services enhance information accessibility and preservation.

Examples include:

- Photocopying services

- Printing and scanning
- Audio-visual services
- Multimedia viewing stations

**LESSON TWO**

**MANAGEMENT PRINCIPLES AND APPLICATIONS TO INFORMATION SYSTEMS**

Management is a fundamental concept in Library and Information Science because Libraries and Information Centres operate as organized systems that require planning, coordination, leadership, and control. Information systems, whether manual or computerized, cannot function effectively without sound management principles guiding their design, implementation, and operation. Management principles provide the framework through which information resources, technologies, staff, and services are coordinated to achieve organizational goals efficiently and effectively.

Management refers to the process of planning, organizing, directing, coordinating, and controlling human, financial, technological, and material resources in order to achieve predetermined objectives.

In the context of information systems, management ensures that information resources are properly created, processed, stored, retrieved, and disseminated to users.

**Concept of Information Systems**

An Information System (IS) is an organized combination of people, hardware, software, data, procedures, and networks designed to collect, process, store, and distribute information for decision-making and service delivery.

**Components of Information Systems**

1. People (librarians, system analysts, users)
2. Hardware (computers, servers, scanners)
3. Software (library management systems)
4. Data/Information resources
5. Procedures and policies
6. Communication networks

**PRINCIPLES OF MANAGEMENT (CLASSICAL MANAGEMENT PRINCIPLES)**

**Planning**

**This** refers to the process of determining organizational objectives and deciding in advance the actions required to achieve them. Planning provides direction for organizational activities and reduces uncertainty by anticipating future needs and challenges. In library and information systems, planning involves setting service goals, developing collection policies, planning automation projects, budgeting for information resources, and designing strategies for effective service delivery. Through planning, managers ensure that available resources are used purposefully and systematically.

**Organising**

This focuses on arranging organizational resources and activities into a structured system that facilitates efficient performance. Organizing involves defining roles, assigning responsibilities, establishing authority relationships, and coordinating tasks among staff members. Within libraries, organizing ensures proper workflow among acquisition, catalog

uing, circulation, reference, and digital services units. Effective organization prevents dup lication of duties, enhances cooperation, and promotes smooth operation of information systems.

**Staffing**

Staffing emphasizes the recruitment, selection, training, and development of competent personnel required for organizational success. Human resources remain the most valuab le asset of any organization, and effective staffing ensures that qualified individuals occ upy appropriate positions. In library environments, staffing includes hiring professional li brarians, information technologists, and support staff, as well as providing continuous p rofessional development to enable adaptation to technological changes and emerging in formation services.

**Directing**,

Directing often referred to as leadership, involves guiding, supervising, motivating, and c ommunicating with employees to ensure that organizational objectives are achieved. Dir ecting ensures that staff members understand their duties and remain committed to insti tutional goals. Library managers apply directing through supervision of daily operations, encouragement of teamwork, effective communication, and motivation of staff to adopt new technologies and improve service quality.

**Controlling**

Controlling is the process of monitoring organizational activities to ensure that actual pe rformance conforms to planned objectives. Control involves setting standards, measurin g performance, identifying deviations, and implementing corrective actions where necess ary. In libraries and information systems, controlling includes evaluating service effective ness, monitoring resource utilization, assessing user satisfaction, and ensuring adherenc e to policies and professional standards.

## MODERN MANAGEMENT PRINCIPLES/APPROACH

### Systems Approach to Management

The systems approach to management views an organization as a unified and interrelated structure composed of various subsystems working together to achieve common objectives. Rather than managing departments or activities in isolation, this approach emphasizes interdependence among organizational components such as people, technology, procedures, resources, and environment. Libraries and Information Centres are considered open systems that continuously interact with users, technological developments, educational institutions, and society at large.

Within information systems management, the systems approach recognizes that hardware, software, databases, communication networks, policies, and human resources must function collectively rather than independently. For example, an automated library system integrates acquisition, cataloguing, circulation, serials control, and online public access catalogues into a single coordinated platform. Any change in one subsystem, such as database structure or software upgrade, directly affects other components of the system. Effective management therefore requires holistic planning, coordination, and continuous monitoring to ensure system balance and efficiency.

### Human Relation Approach

The human relations approach emphasises the importance of people as the central element of organizational success. Unlike earlier management theories that focused primarily on structure and efficiency, this approach recognizes that employee motivation, communication, teamwork, and job satisfaction significantly influence organizational performance. In information systems environments, technology alone cannot guarantee effective service delivery; competent, motivated, and cooperative personnel are essential.

In Libraries and Information Centres, the introduction of information systems often requi

res staff to adapt to new technologies, workflows, and professional roles. The human rel ations approach guides managers to involve staff in decision-making processes, provide adequate training opportunities, and create supportive work environments that encourag e innovation and professional growth. When librarians participate in planning and imple menting information systems, resistance to technological change is reduced, and syste m adoption becomes smoother.

**Total Quality Management (TQM) Approach**

Total Quality Management is a modern management philosophy that focuses on contin uous improvement, service excellence, and customer satisfaction through the involveme nt of all members of an organization. In the context of Library and Information Science, TQM shifts attention from merely providing information resources to ensuring high-quali ty information services that consistently meet or exceed user expectations.

Applying TQM to information systems means that libraries continuously evaluate and re fine their digital services, databases, catalogues, and information delivery processes. Qu ality is not viewed as a one-time achievement but as an ongoing organizational commit ment. Information managers collect feedback from users, analyze system performance, i dentify service gaps, and implement improvements to enhance efficiency and accessibili ty. For instance, libraries may upgrade search interfaces, improve metadata accuracy, red uce system downtime, or expand remote access services based on user feedback and pe rformance evaluation.